OOP Model Practice Paper

Object-Oriented Programming - Demo paper - 2024 edition - 50 marks

> *Self-created practice paper for NoteNest, not an official university paper.*

Section A - Short answers (5 x 4 marks)

Q1. Define class and object with an example. **[4 marks]**

Q2. What is encapsulation? How does it protect an object's data? **[4 marks]**

Q3. Differentiate between method overloading and overriding. **[4 marks]**

Q4. What is the difference between an abstract class and an interface? **[4 marks]**

Q5. Explain the purpose of the finally block. **[4 marks]**

Section B - Long answers (2 x 15 marks)

Q6. Design classes Shape, Circle and Rectangle with an abstract area() method. Show how polymorphism lets you compute the total area of a list of shapes. **[15 marks]**

Q7. Explain checked and unchecked exceptions. Write a program that defines a custom InsufficientBalanceException and uses it in a withdraw() method. **[15 marks]**

Answer hints

- Q3: overloading is resolved at compile time within one class; overriding at run time across a hierarchy.
- Q6: store shapes in a List<Shape> and call area() on each - dynamic dispatch picks the right version.